

Ganesh Gupta

817-822-1131 | ganeshg12ta@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built **AI agent** backends in **Python/FastAPI** and **PostgreSQL**, taking agentic capabilities from prototype to production at **millions of daily transactions** and **99%+ uptime**.
- Built **MCP** tool-use and **function calling** across **GPT, Claude, Llama**, and **Gemini**, tuning model choice per task on quality, latency, and **cost per request**.
- Designed **agent state management** with **Redis checkpointing** so interrupted multi-step workflows resume from the last committed step, making long tool-call chains **reliable**.
- Shipped a deployment-gated **evaluation** framework with automated regression **testing**, blocking **15%** of failed releases before production.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Improved retrieval recall **61% to 89%** on a 2K-query eval set by rebuilding chunking with **hybrid dense-sparse retrieval** and re-ranking.
- Owned **ML infrastructure** end to end, from model deployment to evaluation and debugging, cutting **p99 latency 420ms to 140ms** across **50K+ daily requests**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** for **1M+ users** on **AWS** (EC2, RDS, SQS), sustaining **99.9% uptime** over four years.
- Partnered with **enterprise clients** to turn business processes into technical workflows and custom integrations across 25+ services.
- Standardized failure-handling contracts across services and drove root-cause analysis, cutting production escalations **60%** in one quarter.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent decomposing natural-language questions into dependency-ordered SQL subqueries with hybrid retrieval and re-ranking; deployable as an **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent** incident-diagnosis system with **tool-use** chains that collects evidence from logs and metrics and synthesizes root-cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- LLM evaluation** platform with rubric-based scoring, drift detection, and latency tracking across model versions; **FastAPI**, **PostgreSQL**, regression alerting.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, TypeScript, Java, SQL

Backend: FastAPI, PostgreSQL, REST APIs, Async Services, Redis, Kafka

Agentic AI: AI Agents, Tool Use, LangGraph, OpenAI Agents SDK, MCP, RAG, Evaluation, Guardrails

Cloud & DevOps: AWS, GCP, Kubernetes, Docker, Terraform, CI/CD, Prometheus, Grafana

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, HTML, CSS

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer